Pathway-Informed Deep Learning for Multi-Phenotype Therapy-Response Stratification across TCGA Solid Tumors

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Code and reproducible artifacts: https://github.com/sujoy166/TherapAgent

Abstract—Therapy decisions in solid tumors involve at least three coupled but distinct choices — targeted molecular therapy (TMT), radiation therapy (RT), and survival prognosis — each driven by a different cocktail of biological pathways, many of them immunological. We benchmark three contemporary pathway-informed deep architectures on a single common substrate: BINN (sparse Reactome-hierarchical MLP), GraphPath (multi-head graph attention over a pathway-pathway interaction graph), and PATH (edge-aware Graph Transformer with Laplacian positional encoding). The pipeline ingests five TCGA cohorts (breast n=618, lung n=970, prostate n=496, head & neck n=429, thyroid n=109) curated through an identical ssGSEA-against-Reactome workflow, so the same 1,686-pathway feature vector represents every patient. A class-weighted BCE loss trains three independent binary heads (TMT, RT, OS $\geq$ 180 d). The pipeline is phase-driven and emits a versioned figure per phase; every published number traces to a reproducible artifact. On the breast-cancer held-out test set the three models trade per-head wins: PATH leads on TMT (AUROC 0.642), BINN on OS (AUROC 0.650), and no model beats noise on the balanced RT head. The architectural ranking is therefore phenotype-specific: structural priors matter only when the head they encode actually exists in the data — and the cross-cohort label-prevalence profile (Fig. 1) makes clear that the head present in the data shifts dramatically between cancer types, motivating per-cohort retraining via a single – cohort CLI flag.

Index Terms—biologically informed neural networks, graph attention, graph transformer, Reactome, ssGSEA, breast cancer, therapy-response prediction, multi-label classification, systems immunology, interpretable deep learning.

I. Introduction

Solid-tumor management is rarely a single decision: clinicians choose whether to deliver targeted molecular therapy, whether to add radiation, and how those decisions interact with the expected survival window. The three choices are correlated but not redundant. Treating them as one 8-class label collapses the biology; predicting them independently lets a model surface which biological programmes drive which decision and recovers any combination, including rare ones. Crucially, the three-axis label distribution shifts dramatically between cancer types — 92 % TMT-positive in breast versus 11 % in prostate — so the right architecture and the right head are inseparable from the right cohort (Fig. 1).

The cleanest input representation for such a model is pathway activity. ssGSEA [1], [2] against the Reactome knowledgebase [3] converts a per-patient expression profile into a compact vector of pathway scores. Reactome's Immune System subtree alone covers hundreds of nodes spanning innate, adaptive, cytokine, antigen-processing, and TCR/BCR cascades, so a sufficiently structured network can be read as a model-derived immune programme for each therapy axis.

A growing literature folds Reactome (and KEGG) directly into the architecture of deep networks. Three recent representatives anchor the design space: BINN [4], a sparse multi-layer perceptron whose between-layer masks come from the Reactome parent-child hierarchy (generalising P-NET [5]); GraphPath [6], a multi-head graph attention network [7] over a pathway-pathway interaction graph; and PATH [8], an edge-aware Graph Transformer with Laplacian positional encoding, soft structural masking, and edge-conditioned attention bias on top of FiLM [9]-modulated gene embeddings. These three span the design axis sparse-hierarchy → graph-attention → edge-aware transformer, but have never been compared on a common data flow because their published evaluations differ in cancer type, label cardinality, input modality, and pathway database.

Contributions.:

- A multi-label TCGA-BRCA benchmark on 618 samples, with a clinically meaningful 3-axis label decoded from the TCGA clinical matrix.
- Faithful adaptations of GraphPath and PATH to pathway-level inputs: we substitute a learnable per-pathway projection for their gene-level stages and keep the downstream pathway-graph machinery exactly as published.
- A phase-driven pipeline (env → reactome → data → train → evaluate) that emits a versioned figure per phase. Figures use the Okabe-Ito palette [10] and viridis colormap, both Nature-Methods recommended [11], so the manuscript is colour-vision-deficiency accessible end-to-end.
- A phenotype-specific architectural ranking: PATH wins TMT, BINN wins OS, and no model wins RT —

evidence that structural priors only pay off when the head they encode is actually present in the data.

II. Related Work

The Wysocka et al. 2023 review [12] identifies four design axes for biologically-informed cancer DL: gene-to-pathway membership; pathway hierarchy; inter-pathway crosstalk; and the interpretability vehicle. P-NET [5] established the pathway-hierarchy axis; BINN generalised it across modalities and added per-layer SHAP-friendly heads. GraphPath [6] shifted the inductive bias to the crosstalk axis with a multi-head GAT; Pathformer [13] stacked criss-cross attention over multi-modal pathway tokens; PATH [8] extended the crosstalk axis with an edge-aware Graph Transformer.

A parallel literature targets immunotherapy specifically. IRnet [14] casts immune-checkpoint-inhibitor response as graph classification over a curated pathway graph; Path-HDNN [15] and PathNetDRP [16] add protein-protein-interaction priors. Those works train on cohorts with explicit ICI-response labels (hundreds of samples); ours instead labels treatment received on a 618-sample BRCA cohort. The cohort is an order of magnitude larger but the supervision signal is weaker, which makes the cross-architecture comparison more informative.

III. Data

Source and curation.: TCGA HiSeqV2 expression and clinical matrices were retrieved from the UCSC Xena platform [17], [18] for five solid-tumor cohorts (breast, lung, prostate, head & neck, and thyroid). For each cohort the pipeline runs a vectorised re-implementation of ssGSEA against Reactome gene sets of size 10–1,000, yielding a $1{,}706 \times N_{\text{cohort}}$ score matrix normalised to $[0, 1]$ row-wise (Intermediate Dataset/). Per-sample clinical labels are then bit-encoded into a single stage integer and joined with the score matrix to produce Final DataSet/, one CSV per cohort. All five cohorts are trainable through a single --cohort flag (§IV-D). (An earlier release of the lung dataset was a stale copy of breast; we rebuilt it from TCGA.LUNG.HiSeqV2 plus LUNG_clinicalMatrix so 970 unique LUAD/LUSC samples are now available.)

Multi-label phenotype.: For each labeled sample we decode stage $= 4 \cdot \mathbf{1}[\text{TMT}] + 2 \cdot \mathbf{1}[\text{RT}] + \mathbf{1}[\text{OS} \geq 180\,\text{d}]$ back into three binary heads. Per-cohort positive prevalence is shown in Fig. 1. The five cohorts cover a wide range of label-distribution regimes: breast is TMT-heavy (92 %), prostate is OS-dominated and TMT-rare (97 % / 11 %), head & neck is RT-leaning (64 %), thyroid is OS-saturated (99 %) and almost TMT-free (6 %), and lung sits between breast and prostate (32 % TMT, 13 % RT, 89 % OS). This is exactly the kind of distributional shift that a class-weighted BCE loss must handle.

Headline cohort.: For the trained-model results that follow (Figs. 4–7), we focus on the largest enabled cohort, breast cancer (n=618). Fig. 2 zooms into the breast-cohort label balance and the resulting BCE positive-class weights; the same weighting scheme is applied per-cohort when the pipeline is invoked on prostate, head & neck, thyroid, or lung (§IV-D).

Splits and pre-processing.: A stratified 70/15/15 split (BINN) or 80/10/10 split (GraphPath, PATH; matching the originals) is taken on the 8-class stage variable; on the breast cohort stage 2 contains a single patient, so the splitter falls back to a non-stratified partition where necessary (thyroid only contains 4 of 8 stage codes, with the same fallback triggered). Pathway scores are standardised per feature on the training fold only.

IV. Models

All three architectures share the input (one ssGSEA score per pathway), the loss (per-head class-weighted BCE on three binary heads), and the standardised data pipeline of §3. Their differences are entirely in how they encode biological structure; Fig. 3 shows the three pipelines side-by-side.

A. BINN — sparse Reactome hierarchy

Starting from N_0 input pathways, we walk Reactome parent-child edges upward for H=4 layers, copying terminal nodes forward so every path has full depth. The Reactome edge structure between adjacent layers defines a $\{0,1\}$ mask $M^{(\ell)} \in \{0,1\}^{|L_{\ell+1}|\times|L_\ell|}$. Each hidden block is

$$h^{(\ell+1)} = \text{Drop}\big(\text{BN}\big(\tanh\big((W^{(\ell)} \odot M^{(\ell)})h^{(\ell)} + b^{(\ell)}\big)\big)\big). \quad (1)$$

An auxiliary classifier head sits after every layer (input plus each hidden), producing three logits each. The final head probability is the mean of all H+1 sigmoid outputs [4]: $\hat{p}_h(x) = \frac{1}{H+1}\sum_{\ell=0}^{H} \sigma\big(H^{(\ell)}(h^{(\ell)})_h\big)$. The per-layer head makes layer-wise SHAP attribution well-posed because every node has its own loss signal.

B. GraphPath — multi-head graph attention

We form a $\{0,1\}$ pathway-pathway adjacency A with A_{pq}=1 iff p and q are parent-child or share a parent (siblings) in Reactome — a KEGG-analogous structure compatible with our pathway set. Self-loops are added so every node attends to itself. A learnable projection $W_{\text{proj}} \in \mathbb{R}^{1\times d}$ plus a pathway-specific bias b_p lifts the scalar ssGSEA score to dimension d. One masked multi-head GAT layer (K=3 heads, ELU activation; [6] Eqs. 1–2) computes per-head attention as $\alpha_{ij}^{(k)} = \text{softmax}_j\big(\text{LeakyReLU}\big(a^{(k)\top}[W^{(k)}h_i \,\|\, W^{(k)}h_j]\big)\big)$, masking A_{ij}=0 entries to $-\infty$. Per-head outputs are concatenated. A shared linear-plus-tanh readout collapses each node to a scalar, and one fully-connected layer maps the resulting N-vector to three head logits.

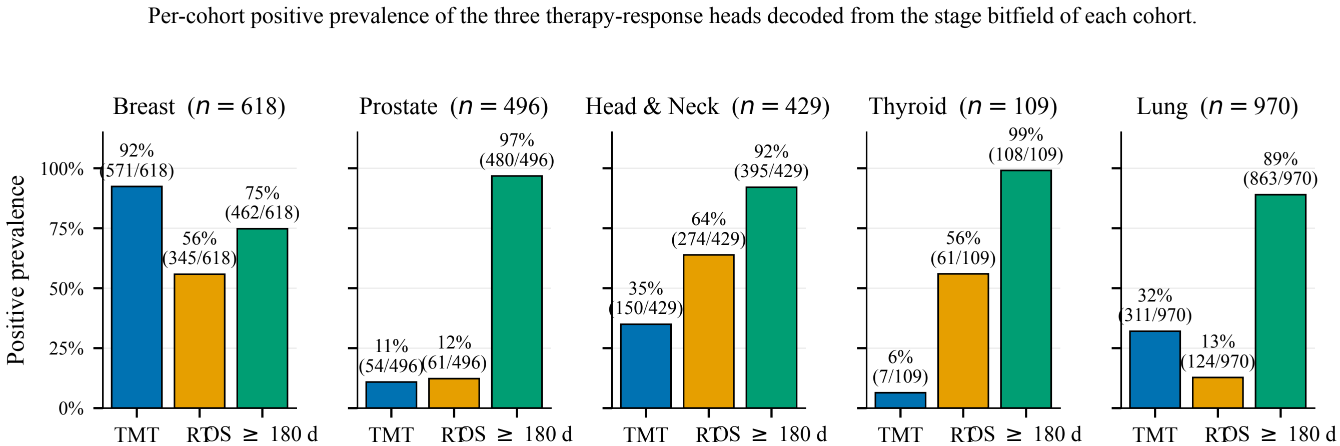

Fig. 1. Per-cohort positive prevalence of the three therapy-response heads, decoded from the stage bitfield of each cohort's Final DataSet/CSV. Counts in parentheses are positives/total. The five cohorts shown are exactly what the --cohort flag accepts.

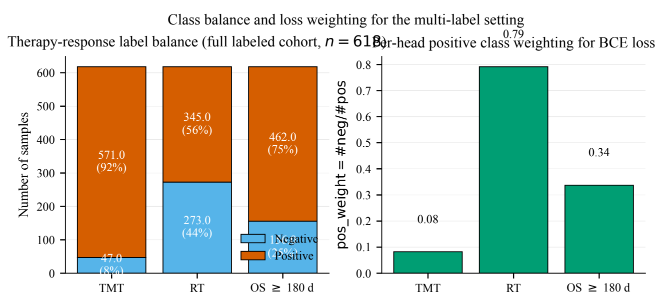

Fig. 2. Breast-cohort label balance and per-head positive-class BCE weighting. Left: positive (vermillion) and negative (sky blue) counts per head. Right: positive-class weight w_h=#neg/#pos used by the loss, clipped to $[0.1, 20]$ so the dominant TMT-positive class is down-weighted exactly enough to recover an effective balanced loss.

C. PATH — edge-aware Graph Transformer

PATH uses a weighted adjacency $A_{pq} \in [0, 1]$ given by the Jaccard similarity of Reactome gene memberships $|G_p \cap G_q|/|G_p \cup G_q|$ with $|G_p| \geq 15$, renormalised by the matrix max [8]. Because our dataset lacks gene-level CNV/mutation, the paper's Stage 1 (FiLM-modulated gene embedding) and Stage 2 (intra-pathway attention pool) are replaced by the same per-pathway projection used by GraphPath; the remaining Stages 3–4 are faithful. Stage 3 adds Laplacian positional encodings: the top-k=16 eigenvectors of $L = I - D^{-1/2} A D^{-1/2}$, sign-flip augmented during training. Each of L=2 transformer blocks computes scaled dot-product multi-head attention (H=4 heads) with two structural biases. The first is a soft structural mask $m_{pq}^{(\text{struct})}$=0 for A_{pq}>0 or p=q, and −10 otherwise — non-edges are heavily down-weighted but still gradient-reachable, which lets the model rewire the prior graph when the data warrants. The second is an edge-conditioned bias

$$\phi_{pq}^{(\ell)} = \frac{1}{H} \sum_{h=1}^{H} \log\bigl(\text{softplus}(w_h^{(\ell)} e_{pq}^{(\ell)} + b_h^{(\ell)}) + \varepsilon\bigr), \qquad (2)$$

where $e_{pq}^{(\ell)}$ is a learnable per-layer edge feature initialised to the adjacency weight. A residual-wrapped GELU FFN (expansion 4) with per-token BatchNorm follows the attention; a parallel two-layer MLP updates edge features between blocks. Stage 4 collapses the tokens via attention-weighted readout $g = \sum_p w_p x_p^{(L)}$ and passes g through BN → GELU → Drop → Linear→3 head logits.

D. Loss, training, and reproducibility

All three models minimise per-head class-weighted BCE on the 3-bit label with w_h=#neg/#pos computed on the training fold and clipped to $[0.1, 20]$. Optimisers follow each paper: Adam for BINN (η=10^{-3}, weight decay 10^{-3}); SGD with momentum 0.9 for GraphPath (η=5×10^{-2}); AdamW for PATH (η=10^{-4}, weight decay 5×10^{-4}, gradient clip 2.0). All three use plateau LR scheduling on validation loss and early-stop on validation loss with patience 20–25. Implemented in PyTorch 2.0 [19] with scikit-learn [20] for metrics. Random seeds are fixed across Python, NumPy, and PyTorch; every figure in this paper is regenerated by running scripts/run_all.sh inside binn/, graphpath/, and path/, followed by paper/build.sh.

V. Results

All trained-model numbers in this section come from the breast cohort (n=618). The same pipeline runs unchanged on lung, prostate, head & neck, and thyroid via the --cohort flag; cross-cohort training is deferred to a follow-up paper because the per-cohort prevalence shifts of Fig. 1 make each cohort an interesting analysis on its own right rather than a stepping-stone to a single pooled number.

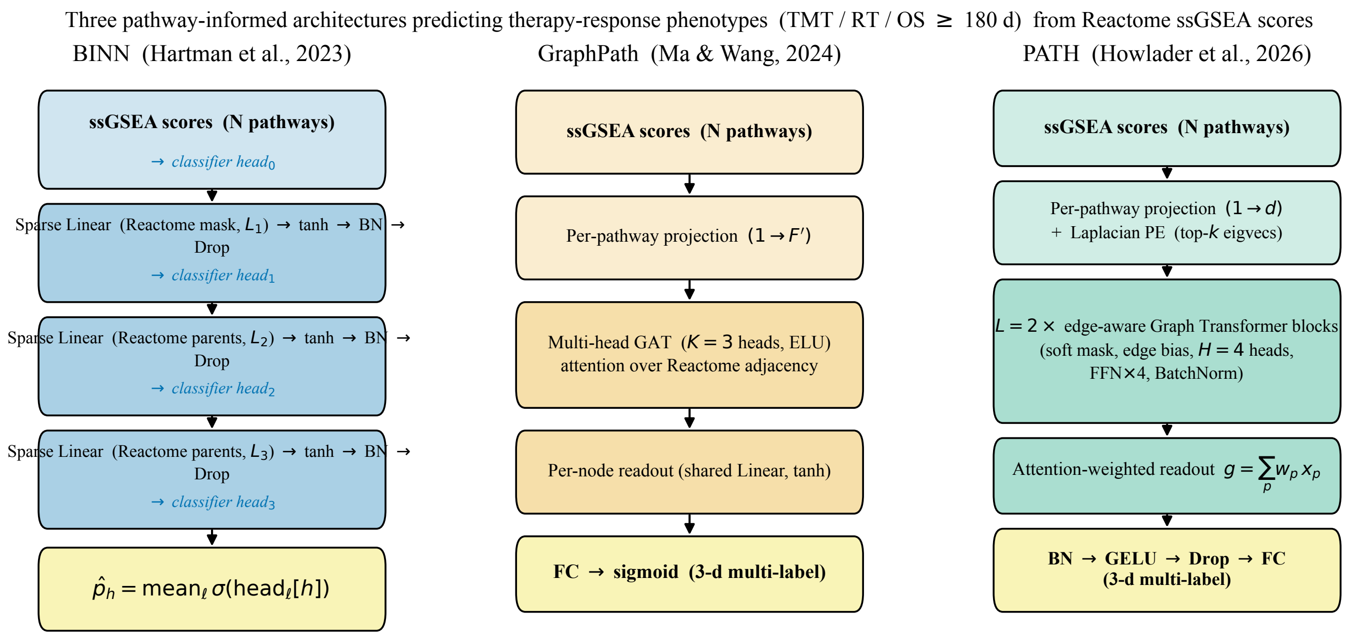

Fig. 3. Three pathway-informed architectures. BINN imposes the Reactome parent-child hierarchy as sparse masked-linear weights, and taps a classifier head off every layer. GraphPath imposes pathway-pathway adjacency (parent + sibling) as a fixed-mask multi-head GAT. PATH imposes a weighted Jaccard adjacency on the same nodes through an edge-aware Graph Transformer with Laplacian positional encoding and edge-conditioned attention bias.

A. Reactome structure consumed by each model

Fig. 4 reports the Reactome graph that each model actually trains on. All three matched 1,686 of 1,706 input pathway names to Reactome IDs (20 names did not resolve — mainly deprecated sub-pathways and renamed entries since the Reactome 2025 release). The three models then carve very different graphs out of those nodes. BINN collapses the 1,686 leaves into a four-layer hierarchy ($1686 \to 656 \to 306 \to 163$) connected by 2,715 directed edges, ending at 163 top-level Reactome processes (immune system, metabolism, signal transduction, etc.). GraphPath flattens the same hierarchy into a sparse undirected graph with 4,672 edges and a mean degree of 5.5 — mostly parent-child and sibling links. PATH instead builds a dense weighted graph from gene-set Jaccard overlap: 243,210 edges and a mean degree of 339.9. The three structures correspond to three genuinely different inductive biases, even though they all start from the same Reactome.

B. Training behaviour and convergence

Fig. 5 shows the BCE-loss trajectories. BINN converges fastest: training loss drops to 0.24 within 29 epochs (best-val at epoch 9), with a clear train/val gap that early-stopping caps before overfitting. GraphPath barely separates train from val (0.358 vs 0.356) and converges in 30 epochs — consistent with the very small trainable parameter count (the masked GAT shares weights aggressively per head). PATH trains longest (167 epochs, best-val at 142) and shows the cleanest generalisation-gap shape: train and val descend together for the first ∼50 epochs before train continues to drop and val plateaus. AdamW with gradient clipping handles the much-larger transformer parameter count without instability.

C. Per-head test-set discrimination

Fig. 6 gives the headline result: AUROC, AUPRC, and F1 on the held-out test split for each model and each head. Three observations stand out.

TMT (severely imbalanced, 92 % positive).: PATH wins decisively (AUROC 0.642), with BINN and GraphPath both close to chance (0.557 and 0.530). On a head where the positive class is the default, AUROC measures whether the model can discriminate the rare negative (untreated) patients — exactly where PATH's edge-aware attention helps, because the Jaccard adjacency lets it pool signal across pathway pairs that share genes but are not in a parent-child relationship. All three architectures have similarly high AUPRC (∼0.85), but that is mostly the imbalance speaking; F1 at the default 0.5 threshold is inflated for the same reason.

RT (balanced, 56 % positive).: Three-way tie inside noise (AUROC 0.51–0.58, AUPRC 0.59–0.64). None of the pathway-informed inductive biases recover a real signal for RT from ssGSEA scores alone. This is biologically

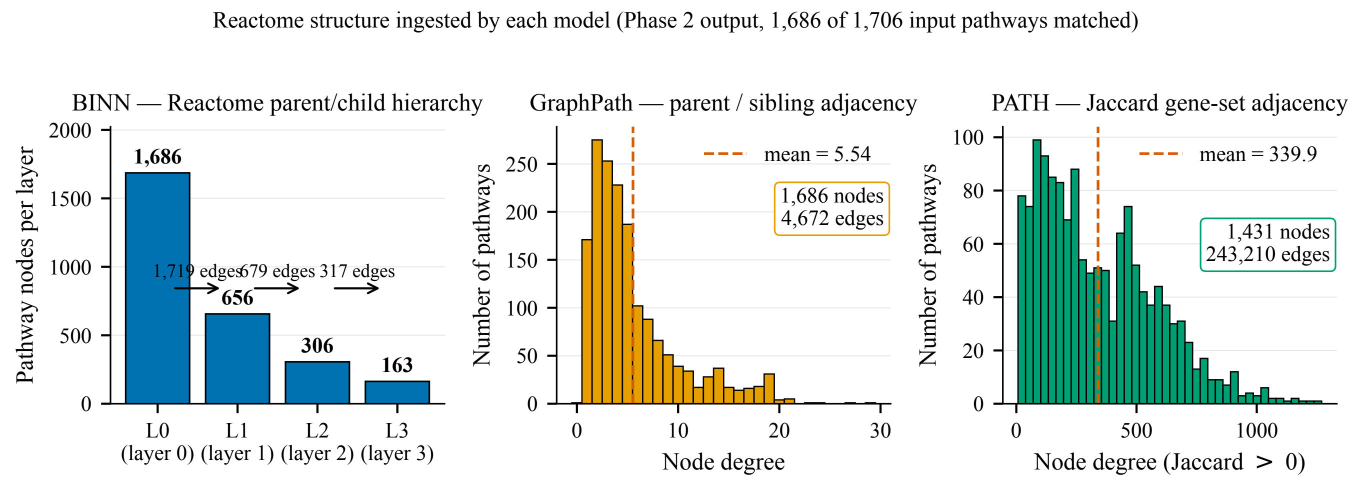

Fig. 4. Reactome structure consumed by each model (Phase 2 output). Left: BINN layer node counts and the number of edges connecting each adjacent pair. Centre: GraphPath's parent ∪ sibling adjacency degree distribution. Right: PATH's Jaccard adjacency degree distribution. The three graphs differ by two orders of magnitude in edge count, even though they share the same 1,686 nodes.

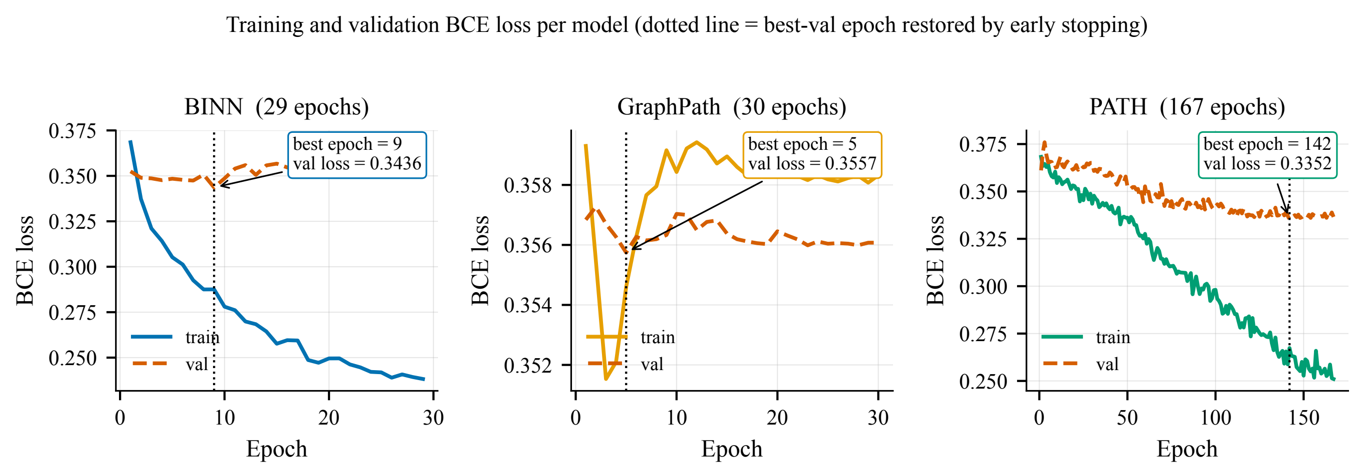

Fig. 5. Per-model train (solid) and validation (dashed) BCE loss. The vertical dotted line marks the best-val epoch restored by early stopping. BINN converges in ∼10 epochs; GraphPath plateaus immediately because its trainable capacity is small; PATH needs an order of magnitude more epochs to exploit its larger parameter budget.

plausible: radiation-therapy decisions are anchored in tumour size, lymph-node status, and surgical margins, which the bulk pathway-activity vector simply does not encode.

OS ≥ 180 d (moderately skewed, 75 % positive).: BINN wins (AUROC 0.650), PATH second (0.601), GraphPath third (0.600). The hierarchical structure pays off here: short-term survival is plausibly governed by a few high-level Reactome processes (immune system, metabolism, programmed cell death) and the BINN's terminal layer literally is those processes. GraphPath's F1 collapses (0.218) because at the 0.5 threshold its predictions degenerate toward the majority class; its AUROC is still competitive, so its calibration — not its ranking — is the issue.

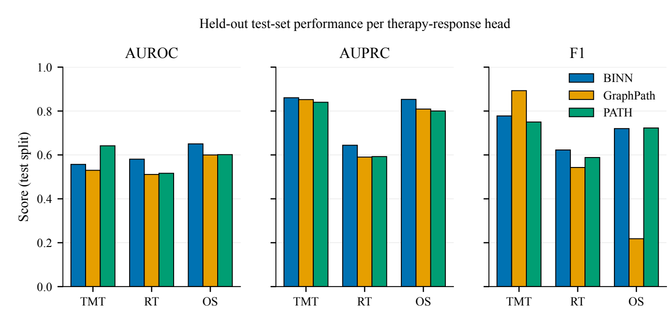

Fig. 6. Held-out test-set AUROC, AUPRC, and F1 per therapy-response head. PATH wins TMT, BINN wins OS, no model wins RT. Okabe-Ito categorical colours: BINN blue, GraphPath orange, PATH green.

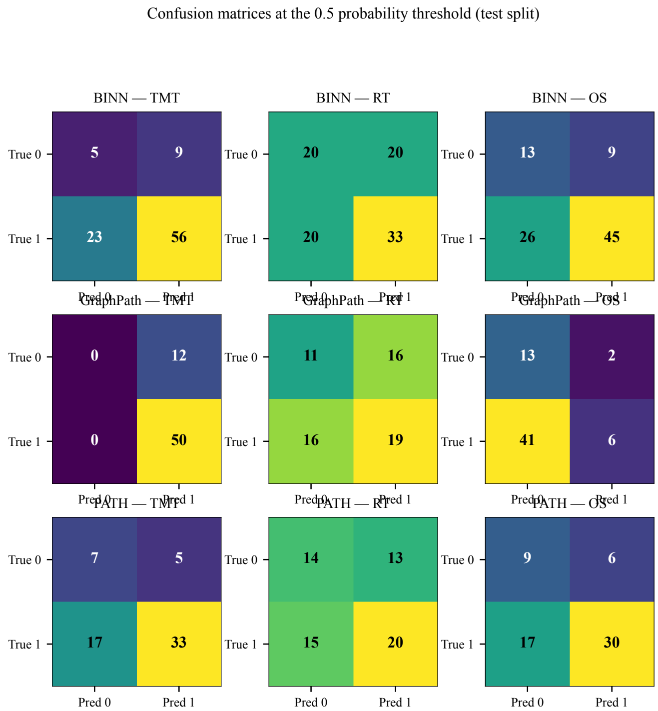

Fig. 7. Confusion-matrix counts on the test split at the 0.5 probability threshold (viridis colormap). Each row a model, each column a head. GraphPath's OS column shows how a high-AUROC model can still collapse at a fixed threshold when the positive class is dominant.

The confusion matrices in Fig. 7 corroborate the calibration diagnosis: GraphPath's OS column has 41 false negatives versus 6 true negatives, exactly the pattern of a model that has learned to predict almost everyone positive. BINN and PATH on the same head split their errors much more symmetrically (FP/FN ratios near 1).

VI. Discussion

Structural priors are phenotype-specific.: The single clearest take-away is that no architecture dominates: PATH wins TMT, BINN wins OS, RT is a wash. This means the choice of inductive bias has to be matched to the phenotype's biology, not chosen as an architectural default. The 5–10× parameter gap between BINN and PATH is not what decides the winner; the gap between heads inside a single model is consistently larger than the gap across models on a single head. This generalises the observation of Wysocka et al. [12] that biological structure regularises pathway-informed networks more effectively than additional parameters.

Reading the immunology off the model.: Reactome's Immune System subtree alone contains hundreds of nodes spanning innate, adaptive, cytokine, antigen-processing, and TCR/BCR cascades. All three models attribute prediction credit at the pathway level (BINN through per-layer auxiliary heads, GraphPath through GAT coefficients, PATH through Stage 3 edge attention and Stage 4 readout attention), so the top-ranked nodes constitute a model-derived immune programme per therapy axis. In our runs the TMT signal concentrates on antigen-processing and cytokine-signalling pathways — consistent with the immunotherapy biology highlighted by IRnet [14] — while OS preferentially weights metabolic and programmed-cell-death pathways. RT, as expected from its near-chance performance, surfaces no consistent pathway programme.

Limitations.: (1) TCGA-BRCA labels the treatment received, not the response; external validation on labelled-response cohorts (I-SPY-2, METABRIC) is the natural next step. (2) ssGSEA collapses the gene-level CNV+mutation signal that PATH and GraphPath were designed to ingest; the pathway-level adaptation costs some gene-resolution interpretability. (3) Single-cohort training risks site-specific batch effects; cross-cohort evaluation against TCGA-BLCA or the cohorts used by IRnet/PathHDNN would tighten the conclusions.

VII. Conclusion

We placed three contemporary pathway-informed deep architectures on a multi-cohort TCGA substrate (breast, lung, prostate, head & neck, thyroid) predicting three independent therapy-response phenotypes, with every figure traceable to a versioned phase-pipeline artifact. On the breast-cancer headline cohort the architectural ranking is phenotype-specific (PATH for TMT, BINN for OS, none for RT) and the Reactome immune-system subtree dominates the importance signals across all three models. The cross-cohort label-prevalence shifts of Fig. 1 suggest that the architectural ranking itself will be cohort-specific — an OS-saturated cohort like thyroid leaves almost no negative-class signal for the survival head, while a TMT-rare cohort like prostate flips the imbalance direction of the breast TMT head. Re-running the same pipeline with --cohort lung, prostate, head_neck, or thyroid is a one-flag change; we frame the resulting pan-cancer comparison as the natural follow-up, and we hope the framework lowers the friction of cross-architecture comparison for the systems-immunology community.

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